

Fish intake and breastfeeding time are associated with serum concentrations of organochlorines in a Swedish population.



Persistent organic pollutants (POPs) exert harmful effects on cognitive, endocrine and immune functions and bioaccumulate in the environment and human tissues. The aim of this study was to investigate the body burden of several POPs in the adult population (n=246) and their association to diet and other lifestyle factors in a Swedish national survey. Serum concentrations of several polychlorinated biphenyls (PCBs), and the pesticides hexachlorobenzene (HCB), β -hexachlorocyclohexane (β -HCH), chlordane compounds and dichlorodiphenyldichloroethylene (DDE) were determined by liquid-liquid extraction, silica column cleanup and gas chromatography high resolution mass spectrometry. Diet was assessed using 4-day food records and complementary dietary and lifestyle factors by questionnaire. Fish intake was additionally assessed by plasma fatty acid composition. Clustering of the compounds revealed that PCBs were separated into two clusters, one including low-chlorinated PCB 28 and 52, and the other high-chlorinated mono- and di-ortho PCBs, suggesting similarities and dissimilarities in exposure sources and possibly also toxicokinetics. Men had 24% and 32% higher levels of PCB 138-180 and chlordane compounds, respectively, compared with women. This may partly be explained by elimination of the POPs among women reporting a history of breastfeeding. The proportion of very long-chain n-3 fatty acids in plasma were positively correlated with the pollutants: $r=0.24$ (PCB 28), $r=0.33$ (PCB 118), $r=0.35$ (PCB 138-180), $r=0.29$ (HCB), $r=0.18$ (β -HCH), $r=0.34$ (chlordane compounds), $r=0.34$ (p,p'-DDE), $p \leq 0.005$. Individuals consuming fatty Baltic fish ≥ 1 time per months had 45% higher serum levels of PCB 118 compared with non-consumers. Levels of PCB 28 were associated with the age of the residential building. To conclude, the population-distributed approach of surveying dietary habits, lifestyle factors and POP body burdens, made it possible to identify personal characteristics associated with the POP body burdens in Sweden. Copyright © 2012 Elsevier Ltd. All rights reserved.